Zener Voltage Regulators

250 mW SOD-923 Surface Mount

NZ9F2V4T5G, SZNZ9F2V4T5G SERIES

This series of Zener diodes is packaged in a SOD-923 surface mount package. They are designed to provide voltage regulation protection and are especially attractive in situations where space is at a premium. They are well suited for applications such as cellular phones, hand held portables, and high density PC boards.

Specification Features:

- Standard Zener Breakdown Voltage Range – 2.4 V to 24 V
- Steady State Power Rating of 250 mW
- Small Body Outline Dimensions: 0.039″ x 0.024″ (1.00 mm x 0.60 mm)
- Low Body Height: 0.016″ (0.40 mm)
- ESD Rating of Class 3 (>16 kV) per Human Body Model
- SZ Prefix for Automotive and Other Applications Requiring Unique Site and Control Change Requirements; AEC-Q101 Qualified and PPAP Capable
- These Devices are Pb-Free, Halogen Free/BFR Free and are RoHS Compliant

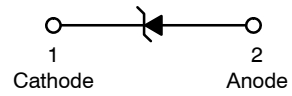
Mechanical Characteristics:

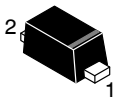
CASE: Void-free, transfer-molded, thermosetting plastic
Epoxy Meets UL 94 V-0
LEAD FINISH: 100% Matte Sn (Tin)
MOUNTING POSITION: Any
QUALIFIED MAX REFLOW TEMPERATURE: 260 °C
Device Meets MSL 1 Requirements

MAXIMUM RATINGS

Rating	Symbol	Max	Unit
Total Device Dissipation FR-5 Board, (Note 1) @ T_A = 25 °C Derate above 25 °C	P_D	250 2.0	mW mW/°C
Thermal Resistance from Junction-to-Ambient	$R_{\theta JA}$	500	°C/W
Junction and Storage Temperature Range	T_J, T_{stg}	−65 to +150	°C

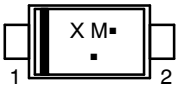
Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.
1. FR-4 Minimum Pad.





SOD-923
CASE 514AB

MARKING DIAGRAM



X = Specific Device Code
M = Month Code
▪ = Pb-Free Package
(Note: Microdot may be in either location)

ORDERING INFORMATION

Device	Package	Shipping†
NZ9FxxxxT5G, SZNZ9FxxxxT5G	SOD-923 (Pb-Free)	8000 / Tape & Reel

† For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specification Brochure, BRD8011/D.

NOTE: Some of the devices on this data sheet ha **DISCONTINUED**. Please refer to the table on page 3.

DEVICE MARKING INFORMATION

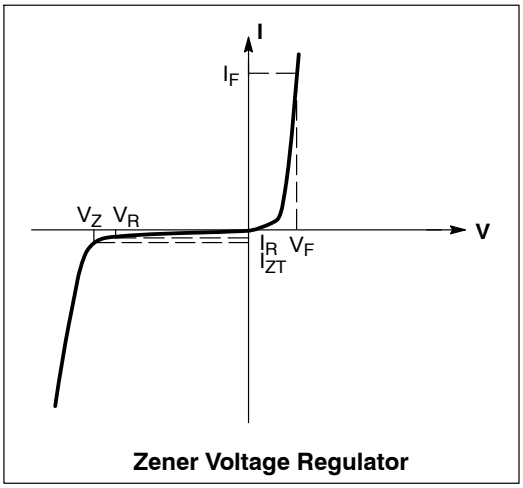
See specific marking information in the device marking column of the Electrical Characteristics tables starting on page 3 of this data sheet.

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ELECTRICAL CHARACTERISTICS

(T_A = 25 °C unless otherwise noted,
V_F = 0.9 V Max. @ I_F = 10 mA for all types)

Symbol	Parameter
V_Z	Reverse Zener Voltage @ I_{ZT}
I_{ZT}	Reverse Current
Z_{ZT}	Maximum Zener Impedance @ I_{ZT}
I_{ZK}	Reverse Current
Z_{ZK}	Maximum Zener Impedance @ I_{ZK}
I_R	Reverse Leakage Current @ V_R
V_R	Reverse Voltage
I_F	Forward Current
V_F	Forward Voltage @ I_F
ΘV_Z	Maximum Temperature Coefficient of V_Z
C	Max. Capacitance @V_R = 0 and f = 1 MHz



Zener Voltage Regulator

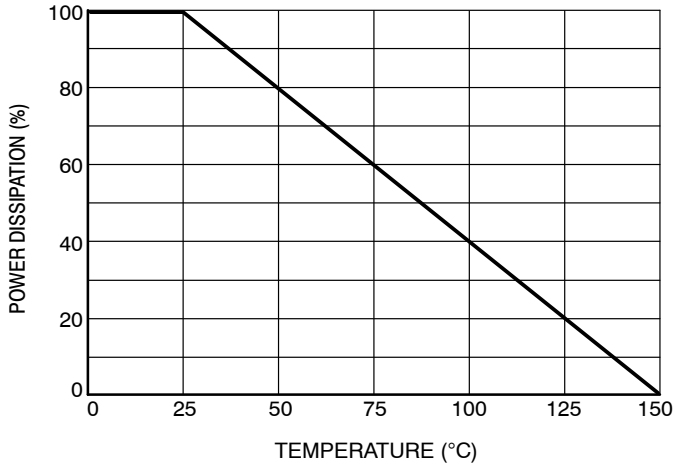


Figure 1. Steady State Power Derating

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ELECTRICAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted, V_F = 0.9 V Max. @ I_F = 10 mA for all types)

Device***	Device Marking	Zener Voltage (Note 1) V_Z (Volts) Min	V_Z (Volts) Max	@ I_{ZT} mA	Zener Impedance Z_{ZT} @ I_{ZT} Ω	Z_{ZK} @ I_{ZK} Ω	I_{ZK} mA	Leakage Current I_R @ V_R µA	V_R Volts	ΘV_Z (mV/k) @ I_{ZT} Min	Max	C @ V_R = 0 f = 1 MHz pF
SZ, NZ9F2V4T5G	J	2.28	2.52	5	100	1000	1	50	1	−3.5	0	210
SZNZ9F2V7T5G	E**	2.57	2.84	5	100	1000	1	20	1	−3.5	0	210
SZNZ9F3V0T5G	T**	2.85	3.15	5	100	1000	1	10	1	−3.5	0	210
SZ, NZ9F3V3T5G	Q	3.14	3.47	5	100	1000	1	10	1	−3.5	0	210
SZNZ9F3V6T5G	3**	3.42	3.78	5	100	1000	1	10	1	−3.5	0	210
SZ, NZ9F3V9T5G	V**	3.71	4.10	5	100	1000	1	5	1	−3.5	−2.5	210
SZNZ9F4V3T5G	Y**	4.09	4.52	5	100	1000	1	5	1	−3.5	0	210
SZ, NZ9F4V7T5G	3	4.47	4.94	5	100	800	0.5	2	1	−3.5	0.2	150
SZ, NZ9F5V1T5G	4	4.85	5.36	5	80	500	0.5	2	1.5	−2.7	1.2	130
SZ, NZ9F5V6T5G	5	5.32	5.88	5	60	200	0.5	1	2.5	−2.0	2.5	115
SZ, NZ9F6V2T5G	6	5.89	6.51	5	60	100	0.5	1	3	0.4	3.7	110
SZ, NZ9F6V8T5G	A*	6.46	7.14	5	40	60	0.5	0.5	3.5	1.2	4.5	105
SZNZ9F7V5T5G	D*	7.13	7.88	5	30	60	0.5	0.5	4	2.5	5.3	100
SZNZ9F8V2T5G	E*	7.79	8.61	5	30	60	0.5	0.5	5	3.2	6.2	90
SZNZ9F9V1T5G	F*	8.65	9.56	5	30	60	0.5	0.5	6	3.8	7	80
SZ, NZ9F10VT5G	J*	9.50	10.50	5	30	60	0.5	0.1	7	4.5	8	80
SZ, NZ9F11VT5G	K*	10.45	11.55	5	30	60	0.5	0.1	8	5.4	9	80
NZ9F12VT5G	L*	11.40	12.60	5	30	80	0.5	0.1	9	6	10	80
SZNZ9F13VT5G	P*	12.35	13.65	5	37	80	0.5	0.1	10	7	11	75
SZ, NZ9F15VT5G	Q*	14.25	15.75	5	42	80	0.5	0.1	11	9.2	13	70
SZ, NZ9F16VT5G	R*	15.20	16.80	5	50	80	0.5	0.1	12	10.4	14	65
SZ, NZ9F18VT5G	T*	17.10	18.90	5	50	80	0.5	0.1	14	12.4	16	60
SZ, NZ9F20VT5G	V*	19.00	21.00	5	55	100	0.5	0.1	15.4	14.4	18	55
SZ, NZ9F22VT5G	Y*	20.90	23.10	5	55	100	0.5	0.1	16.8	15.4	20	55
SZ, NZ9F24VT5G	F	22.80	25.20	5	70	120	0.5	0.1	18.9	16.8	22	50

DISCONTINUED (Note 2)

NZ9F2V7T5G	E**	2.57	2.84	5	100	1000	1	20	1	−3.5	0	210
NZ9F3V0T5G	T**	2.85	3.15	5	100	1000	1	10	1	−3.5	0	210
NZ9F3V6T5G	3**	3.42	3.78	5	100	1000	1	10	1	−3.5	0	210
NZ9F4V3T5G	Y**	4.09	4.52	5	100	1000	1	5	1	−3.5	0	210
NZ9F7V5T5G	D*	7.13	7.88	5	30	60	0.5	0.5	4	2.5	5.3	100
NZ9F8V2T5G	E*	7.79	8.61	5	30	60	0.5	0.5	5	3.2	6.2	90
NZ9F9V1T5G	F*	8.65	9.56	5	30	60	0.5	0.5	6	3.8	7	80
SZNZ9F12VT5G	L*	11.40	12.60	5	30	80	0.5	0.1	9	6	10	80
NZ9F13VT5G	P*	12.35	13.65	5	37	80	0.5	0.1	10	7	11	75

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

* Rotated 90°.

* *Rotated 270°.

* **SZ Prefix for Automotive and Other Applications Requiring Unique Site and Control Change Requirements; AEC-Q101 Qualified and PPAP Capable.

1. Zener voltage is measured with a pulse test current I_Z at an ambient temperature of 25 °C.
2. **DISCONTINUED:** These devices are not available. Please contact your **onsemi** representative for information. The most current information on these devices may be available on www.onsemi.com.

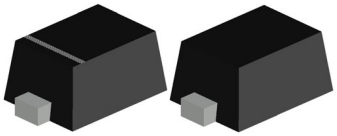
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REVISION HISTORY

Revision	Description of Changes	Date
5	Rebranded the Data Sheet to **onsemi** format. NZ9F2V7T5G, NZ9F3V0T5G, NZ9F3V6T5G, NZ9F4V3T5G, NZ9F7V5T5G, NZ9F8V2T5G, NZ9F9V1T5G, SZNZ9F12VT5G, NZ9F13VT5G OPNs Marked as Discontinued	09/30/2025

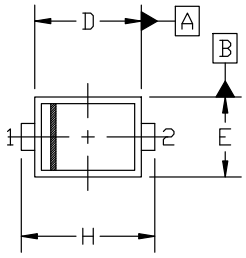
This document has undergone updates prior to the inclusion of this revision history table. The changes tracked here only reflect updates made on the noted approval dates.

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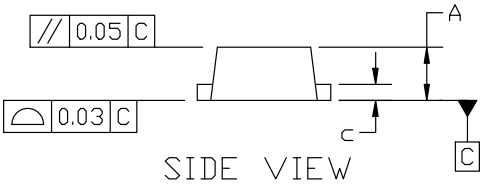


SOD−923 0.80x0.60x0.37
CASE 514AB
ISSUE E

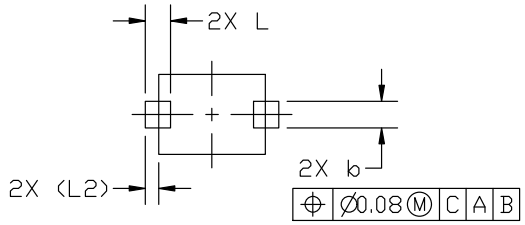
DATE 08 FEB 2024



TOP VIEW



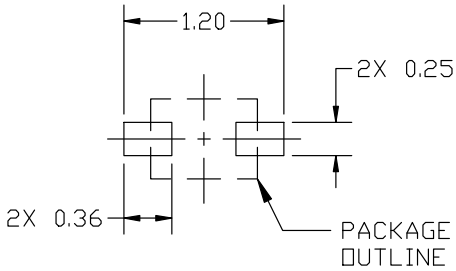
SIDE VIEW



BOTTOM VIEW

NOTES:
1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 2018.
2. CONTROLLING DIMENSION: MILLIMETERS.
3. MAXIMUM LEAD THICKNESS INCLUDES LEAD FINISH. MINIMUM LEAD THICKNESS IS THE MINIMUM THICKNESS OF BASE MATERIAL.
4. DIMENSIONS D AND E DO NOT INCLUDE MOLD FLASH, PROTRUSIONS, OR GATE BURRS.
5. DIMENSION L WILL NOT EXCEED 0.30mm.

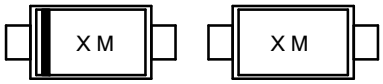
MILLIMETERS			
DIM	MIN.	NOM.	MAX.
A	0.34	0.37	0.40
b	0.15	0.20	0.25
c	0.07	0.12	0.17
D	0.75	0.80	0.85
E	0.55	0.60	0.65
H	0.95	1.00	1.05
L	0.19 REF		
L2	0.05	0.10	0.15



RECOMMENDED MOUNTING FOOTPRINT

*For additional information on our Pb-Free strategy and soldering details, please download the ON Semiconductor Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

GENERIC MARKING DIAGRAM*



STYLE 1 STYLE 2

X = Specific Device Code
M = Date Code

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, “G” or microdot “▪”, may or may not be present. Some products may not follow the Generic Marking.

STYLE 1:
PIN 1. CATHODE (POLARITY BAND)
2. ANODE

STYLE 2:
NO POLARITY

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DESCRIPTION:	**SOD−923 0.80x0.60x0.37**		**PAGE 1 OF 1**

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